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Classical Bit Versus Quantum Qubit

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Assessment 1

1. Which of the following statements accurately describes quantum entanglement? **1 / 1 point**

- [ ] Quantum entanglement is a concept within classical mechanics
- [ ] Entanglement can only occur at absolute zero temperature
- [x] Entangled particles remain connected and share information regardless of the distance between them

Correct
Well done! This is a key characteristic of quantum entanglement.

- [x] Particles become interconnected such that the state of one cannot be described independently of the state of the other

Correct
Correct! Quantum entanglement implies that the state of one particle is directly related to the state of another, no matter the distance between them.

2. Which principle explains that a subatomic particle can exhibit both wave and particle characteristics? **1 / 1 point**

- [ ] Classical mechanics
- [ ] Superposition
- [ ] Quantum entanglement
- [x] Wave-particle duality

Correct
Correct! The principle that a subatomic particle can exhibit both wave and particle characteristics is known as wave-particle duality.

3. What is a fundamental challenge of controlling qubits at room temperature? **1 / 1 point**

- [ ] Qubits need vacuum conditions.
- [ ] Magnetic fields are too weak.
- [x] Thermal noise disrupts quantum states.
- [ ] Qubits require a significant amount of light.

Correct
Correct! Thermal noise can easily disrupt the delicate quantum states of qubits, making it difficult to control them precisely at room temperature.

4. Which features differentiate qubits from classical bits? **1 / 1 point**

- [ ] Ability to exist at room temperature
- [x] Representation on a Bloch sphere

Correct
Correct. Qubits are represented on a Bloch sphere, which illustrates their quantum state, unlike classical bits.

- [ ] Use of transistors for state representation
- [x] Capability of entanglement

Correct
Correct. Qubits can be entangled, creating correlations between their states even at a distance, which is not possible with classical bits.

- [x] Existence in a superposition of states

Correct
Correct. Qubits can exist in superposition, unlike classical bits which can only be in one state at a time.

5. Which of the following are implications of qubits being in superposition? **1 / 1 point**

- [ ] Increased thermal stability
- [x] Parallel processing capabilities

Correct
Correct. Superposition enables parallel processing capabilities, allowing quantum computers to solve complex problems more efficiently.

- [ ] Enhanced magnetic field sensitivity
- [x] Exponential state growth

Correct
Correct. Superposition allows qubits to represent multiple states simultaneously, leading to exponential state growth.

- [ ] Qubits can exist in a vacuum